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What can we learn from the history of gasoline crack spreads?: Long memory, structural breaks and modeling implications

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ABSTRACT

In this paper, we investigate the long-range auto-correlations of crack spreads using a nonparametric method, named detrended moving average (MF-DMA). We find that the auto-correlations display multiscaling behaviors and are dominated by the anti-persistence (mean-reversion) in the long-term. Moreover, the auto-correlations are multifractal, indicating that various small and large fluctuations display different scaling behaviors. Using a technique of rolling windows, we find that some extreme events can drive the degree of anti-persistence and the multifractality (complexity) to rise up. In other words, these events have negative impacts on market efficiency. However, the effects of these events are not alike. We also detect long-range auto-correlations in crack spread volatilities and find a strong persistent behavior and multifractality. Finally, we discuss the modeling implications of the findings on long-range auto-correlated patterns. Our results indicate that ARFIMA-GARCH models can capture the major dynamics of large fluctuations. For small fluctuations, they are misspecified. Interestingly, we find that the strong long-range auto-correlated behaviors do not imply that ARFIMA model which takes long memory into account can outperform random walk model in the sense of out-of-sample prediction. The major reason may be that market complexity exploited in this paper causes the low predictability of ARFIMA model.

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1. Introduction

In oil markets, crack spread is interpreted as the profit margin of refining process. Variations in crack spread contain the information about marginal production costs and reflect the changes in cost structure. Especially, the gasoline crack spread can be simply denoted by the price differential between West Texas Intermediate (WTI) crude oil and unleaded gasoline. Using the following equation, gasoline crack spread is calculated as,

$$CS_t = GL_t - CO_t \,. \tag{1}$$

Where, CS_t is the crack spread of gasoline at time t (dollars per barrel). GL_t and CO_t are the prices of unleaded gasoline and WTI crude oil at time t, respectively (dollars per barrel).

The crack spread interpreted in Eq. (1) denotes an equal-weighted portfolio with a long position of gasoline and a short position of crude oil. It refers to the price relationship between crude oil and gasoline. Oil refiners are the major participants in markets who are exposed to the crack spread and they aim to protect the crack spread. Oil consumers are also frequently exposed to the crack spread and they prefer to the lower level of crack spread. Thus, crack spread derivatives are is favored by most of the participants for hedging.

Since the dynamics of crack spreads are of great importance for oil market participants, whether the history information of crack spread is helpful for predicting its future performance is a crucial issue because it determines the availability of technical analysis. For this motivation, we investigate the long-range auto-correlations of gasoline crack spreads. The motivation for this work also emanates from several existing studies. Evidence in existing literature indicates that crude oil returns and volatilities are long-range auto-correlated (Alvarez-Ramirez et al., 2008; Tabak and Cajueiro, 2007; Wang and Liu, 2010; Wang et al., 2010). We can expect that crack spreads and oil prices have different dynamics. For example, some supply and demand shocks can drive crude oil and gasoline prices to increase or decrease largely. Then, if crude oil and gasoline experience the same price change, the effects of these shocks on the price differentials (crack spreads) may be very minor. However, whether the crack spreads also display meaningful long-range auto-correlated behaviors has not been investigated.

In this framework, we try to answer several questions on the crack spreads as follows. Alvarez-Ramirez et al. (2008) and Wang and Liu (2010) find that dynamics of crude oil markets behave like a system with a high degree of complexity[1] quantified by the strength

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[1] The market complexity origins from many market internal factors such as the behaviors of speculators, noise traders, OPEC and large oil refiners and also from market external factors such as geopolitical events and macroeconomic policy of the United States. In the works of Alvarez-Ramirez et al. (2008) and Wang and Liu (2010), the complexity is investigated by quantifying the difference of scaling behaviors and auto-correlated patterns among various small and large fluctuations.

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of multifractality. That is, various small and large fluctuations of crude oil prices display different scaling behaviors. According to Wang et al.(2010), a high degree of complexity implies the market inefficiency. Thus, crude oil market is not efficient in Fama(1970)'s sense. Do their results still hold for crack spreads? Is the scaling behavior or complexity of crack spreads time-varying? Alvarez-Ramirez et al. (2008), Wang and Liu (2010) and Alvarez-Ramirez et al. (2010) have shown that some supply shocks (e.g., geopolitical events in the Middle East) and demand shocks (e.g., the recent financial crisis) can affect scaling behaviors and complexity of crude oil returns. Do these shocks also have effects on scaling behaviors or complexity of crack spreads? Are these shocks temporary or persistent? If crack spreads and their volatilities are long-range auto-correlated, could conventional models which take into account the long-range correlations well capture the scaling behaviors and complexity of crack spreads? Moreover, if crack spreads are long-range auto-correlated, it implies that autoregressive fractionally integrated moving average (ARFIMA) model which considers the long-range correlations can capture the dynamics of crack spreads better than random walk model (RWM). Does this implication still hold in the sense of out-of-sample prediction?

For the purpose of detecting long-range auto-correlations in crack spreads, we employ a multifractal detrended moving average (MF-DMA) algorithm proposed by Gu and Zhou (2010), which is an extension of DMA initially introduced by Carbone et al. (2004). Gu and Zhou (2010) show that MF-DMA is more robust than the multifractal detrended fluctuation analysis (MF-DFA) employed in Alvarez-Ramirez et al. (2008) and Wang and Liu (2010). An outstanding advantage of MF-DMA over MF-DFA for the empirical analysis is that MF-DMA leads to more controlled behavior of detrended fluctuations for larger time scales (Gu and Zhou, 2010; Wang et al., 2011b).

In this work, we choose the daily spot price data of WTI crude oil and unleaded gasoline. Our data sample covers the period from 1986 to 2011, about 25 years. Brémond et al. (2011) show that after the introduction at the end of the year 2000 of the law modernizing raw materials markets—the Commodity Futures Modernization Act (CFMA), oil prices run in a new regime different from previous decade. Thus, we divide the whole sample period into two sub-periods. Period I spans from January 1986 to December 2000 and Period II spans from January 2000 to July 2011. Then, we investigate the auto-correlated behavior of crack spread series in each of above two sub-periods based on MF-DMA.

Our evidence from MF-DMA indicates that the crack spreads perform a mean-reverting behavior. Moreover, in Period I, a weak mean-reverting behavior can be found at time scales in the order of days to weeks while they are much stronger at time scales in the order of weeks to months, consistent with the multiscaling behavior claimed by Wang and Liu (2010). In Period II, the crack spreads are also mean-reverting but display the property of scaling invariance. That is, the auto-correlated behaviors at small time scales are consistent with those at large time scales. The evidence from multifractal analysis indicates that the auto-correlated behaviors of crack spreads are multifractal in both of two periods. That is, various small and large fluctuations of crack spreads show different scaling behaviors. In other words, the complexity of crack spreads is significant. We also investigate the scaling behaviors of crack spread volatility and find the strong persistence and complexity.

To investigate the evolution of long-range auto-correlations over time, following Tabak and Cajueiro (2007), Alvarez-Ramirez et al. (2008) and Wang and Liu (2010), we employ the technique of rolling window with the fixed length of 1000 business days (about 4 years). The results show that the Hurst exponents are smaller than 0.5 and change over time implying the time-varying mean-reverting behavior. Some extreme events such as two Gulf Wars and the recent financial crisis can introduce large crashes of Hurst exponents, leading to stronger mean-reverting behaviors. Moreover, our results also indicate that not all of the shocks of these events are alike, generally consistent with the results in Kilian (2010). The influence of some events is temporary and that of the some other events is permanent. We analyze the evolution of complexity (multifractality) of crack spread and the scaling behaviors of volatility using the technique of rolling window. We find that the strong persistence of crack spread volatility nearly has no change over time. In contrast to the dynamics of volatility correlations, the degrees of complexity evolve fiercely and vary at higher levels after mid-2003.

Then, we discuss the modeling and forecasting implications of our results on long-range auto-correlations. We model crack spreads using ARFIMA model and model its volatility using several generalized autoregressive conditional heteroskedasticity (GARCH) (Bollerslev, 1986) models. Our results show that ARFIMA-GARCH models can capture the long-range auto-correlations of crack spreads but cannot capture the multiscaling behaviors, generally consistent with the findings in Wang et al. (2011a). Moreover, we find that ARFIMA-GARCH-class models can well capture the complexity (multifractality) caused by large fluctuations but cannot capture that caused by small fluctuations.

Theoretically, for the crack spread series which are strongly long-range auto-correlated, ARFIMA model can capture their dynamics better than random walk model (RWM). However, our empirical evidence indicates that ARFIMA model cannot outperform RWM in the sense of out-of-sample forecasting. The major reason may be that the complexity of the time series which cannot be captured by ARFIMA model leads to the larger prediction bias.

We organize this paper as follows: We perform a general literature review in the next section. In Section 3, we show the methodology of MF-DMA employed in this work. In Section 4, we describe the data and have a preliminary analysis on the data. In Section 5, we show the empirical results based on MF-DMA and some relevant discussions. In the last section, we conclude this paper.

2. A literature review

To the best of our knowledge, the long-range auto-correlations of crack spreads have not been investigated. Thus, in this section, we will review the recent literature on the long-range auto-correlated behaviors in oil markets and the studies on the crack spreads in the following two sub-sections, respectively.

2.1. Long-range auto-correlations in oil markets

Although multitudes of studies contribute to the long-range auto-correlations in financial markets (e.g., Cajueiro and Tabak, 2004a,b; Cajueiro and Tabak, 2005a,b; Lo, 1991; Matteo et al., 2005), to the best of our knowledge, the earliest investigation of long-range auto-correlations in crude oil markets can be found in Alvarez-Ramirez et al. (2002) only. Using multifractal Hurst analysis based on rescaled range analysis (R/S) (Hurst, 1951), Alvarez-Ramirez et al. (2002) find that the major crude oil markets are consistent with the random walk behavior only at time scales in the order of days to weeks. At the other time scales, the long-range auto-correlations in crude oil returns are significant. Employing R/S analysis, Serletis and Andreadis (2004) find the strong persistence and multifractality in WTI crude oil prices and Henry Hub natural gas prices. Elder and Serletis (2008) test for the fractional integrating dynamics in energy futures market using a new semi-parametric wavelet-based estimator, which is superior to the more prevalent GPH estimator and find the evidence that energy prices display long memory (long-range correlations) and that the particular form of long memory is anti-persistence. As an extension, Serletis and Rosenberg (2007) use DMA to re-examine the long-range auto-correlations in energy prices and find the evidence which supports the anti-persistence revealed in Elder and Serletis (2008). Gu et al. (2010) find that WTI and Brent crude oil returns display

different anti-persistence and multifractality in different periods and two Gulf Wars have shocks on the auto-correlated structure of crude oil markets.

The analysis on the whole time series in above literature cannot reveal the evolution of auto-correlated behaviors over time. Following Cajueiro and Tabak (2004a, 2004b, 2005a, 2005b), Tabak and Cajueiro (2007) employ the technique of rolling sample test to investigate the evolution of long-range auto-correlations in crude oil markets. Using R/S analysis, Tabak and Cajueiro (2007) find that crude oil returns and volatilities display time-varying auto-correlated behaviors, which become weaker and weaker implying more and more efficient operation. Alvarez-Ramirez et al. (2008) investigate the auto-correlations of international crude oil prices on the basis of the estimation of Hurst exponent dynamics for returns. Using DFA, Alvarez-Ramirez et al. (2008) find that crude oil returns are not auto-correlated in the long-term implying that the markets are efficient. However, meaningful auto-correlations are present for time horizons smaller than a month and are weaker and weaker. Wang and Liu (2010) extend the work of Tabak and Cajueiro (2007) and Alvarez-Ramirez et al. (2008) by analyzing the time-varying situation of multiscaling auto-correlations and find that the auto-correlations at different small and large time scales are weaker and weaker. Based on the multiscale DFA used in Wang and Liu (2010), Wang et al. (2010) also study the auto-correlations of WTI crude oil volatilities and find that several external shocks could have significant shocks on the auto-correlated behaviors for small time scales only. Using a lagged DFA, Alvarez-Ramirez et al. (2010) further investigate the auto-correlated pattern of crude oil returns and obtain the evidence that the auto-correlations at different lagged orders become weaker and weaker.

2.2. The literature on the crack spread

The recent literature on the crack spread mainly focuses on option pricing, modeling and describing the dynamics of crack spread and some other studies related to crack spread.

The change of crack spread is of great importance for market participants. Thus, there are many studies which contribute to describing and modeling the dynamics of crack spreads. Girma and Mougoue (2002) investigate the relationships between energy futures spread variability, trading volume and interest rate and find that the lagged volume and interest rate can largely explain the variability of futures spread variability. Dunis et al. (2005) model the gasoline crack spread using Higher Order Neutral Networks (HONNs) and discover the profits in excess of those achieved using a standard MultiLayer Perception (MLP). Evans et al. (2006) find that the correlation filter is a promising method of filtering spread trades. In contradiction to Evans et al. (2006), Dunis et al. (2006a) find that the use of unfiltered correlation model generally produce better results than the filtered model. Dunis et al. (2006b) model the crack spread using nonlinear methods. The results show that the crack spread exhibit asymmetric adjustment toward long-run equilibrium level and the Higher Order Neutral Network with the threshold filter models the crack spread best in the sense of out-of-sample prediction.

Spread derivative is a simple multi-asset derivative and occurs in a wide range of financial markets. Some of the recent literatures focus on the crack spread option pricing. For example, Laurence and Wang (2009) derive in closed form distribution free lower bounds and optimal subreplicating strategies for spread options in a one-period static arbitrage setting. Numerical results on real option data of crack spread in Laurence and Wang (2009) indicate that one discrete lower bound can be far and very close to the actual trading price. Herath et al. (2011) investigate the crack spread option pricing based on a copula approach. Based on two developed copula-based option models of crack spread and using a simulation approach, their results indicate that Clayton model is more appropriate than Gumbel copula and standard models in option pricing.

Some of the literatures contribute to the application of crack spread to hedging strategies and forecasting. Haigh and Holt (2002) develop a conceptual model for a trader hedging the crack spread and analyze the hedging effectiveness for crude oil, gasoline and heating oil futures contracts. Haigh and Holt (2002) find that modeling the time-varying hedging ratios via the multivariate GARCH and taking into account the volatility spillover can reduce the uncertainty significantly. Murat and Tokat (2009) investigate the causality relationships between crack spread and crude oil price and find that crack spread can Granger cause oil price both in the short- and long-run. Additionally, Murat and Tokat (2009) forecast the crude oil price movements with crack spreads and find that both crack spread futures and oil futures outperform RWM.

3. Methodology

Since the seminal work of Peters (1991, 1994), Rescaled Range analysis (R/S) is widely used to detect long-rang auto-correlations in financial markets (Cajueiro and Tabak, 2004a, 2004b; Cajueiro and Tabak, 2005a, 2005b). However, Lo (1991) points out that the R/S analysis is sensitive to large variations of observations and may lead to a biased estimation of long-rang auto-correlations in the presence of short-range auto-correlations. Peng et al. (1994) propose a detrended fluctuation analysis (DFA) when they study the correlations of molecular chains in deoxyribonucleic acid (DNA). This method avoids the spurious detection of apparent long-range auto-correlations that are an artifact of patchiness and has become a widely used technique for the determination of long-range auto-correlations (Alvarez-Ramirez et al., 2008; Wang and Liu, 2010). As an extension, Kantelhardt et al. (2002) propose a multifractal DFA which can be used to investigate the multifractality in nonstationary time series.

DFA and MF-DFA remove the possible trends in time series based on the polynomial fitting. Recently, Gu and Zhou (2010) propose a class of multifractal detrended moving average algorithm (MF-DMA) as an extension of DMA initially introduced by Carbone et al. (2004). Different from DFA, this method removes the local trends by subtracting the local means. The simulating results in Gu and Zhou (2010) show that MF-DMA is more robust than MF-DFA. The MF-DMA can be described as follows (Gu and Zhou, 2010):

Step 1. Consider one time series $a(t)$, $t=1,2,\ldots,N$. Where, N is the length of the series. We construct the new sequences of cumulative sums

$$x(t)=\sum_{i=1}^{t} a(i), \tag{2}$$

Step 2. Calculate the moving average function of the series of cumulative sums in a moving window,

$$\tilde{x}(t)=\frac{1}{n}\sum_{k=-\lceil(n-1)\theta\rceil}^{\lceil(n-1)(1-\theta)\rceil} x(t-k), \tag{3}$$

where, n is the window size, $\lfloor x \rfloor$ is the largest integer not larger than x, $\lceil x \rceil$ is the smallest integer not smaller than x, and θ is the position parameter varying from 0 to 1. Hence, the moving average functions of two series consider the past $\lceil(n-1)(1-\theta)\rceil$ data points and the future $\lfloor(n-1)\theta\rfloor$ data points. In Gu and Zhou (2010), they consider three special cases. The first case $\theta=0$ refers to the backward moving average (Xu et al., 2005), in which the moving average functions are calculated from the past $n-1$ data points. The second case $\theta=0.5$ refers to the centered moving average (Xu et al., 2005), in which

the moving average functions are calculated from half past data points and half future data points. The third case $\theta=1$ corresponds to the forward moving average, where functions are calculated from future $n-1$ data points. In this paper, we only consider the first case, the backward moving average, just $\theta=0$.

Step 3. Detrend the series by subtract the moving average functions, and obtain the residual sequence $\varepsilon(i)$ using the equations as,

$$\varepsilon(i)=x(i)-\tilde{x}(i), \tag{4}$$

where, $n-\lfloor(n-1)\theta\rfloor\le i\le N-\lfloor(n-1)\theta\rfloor$.

Step 4. Divide the residual sequences into N_n non-overlapped segments with the equal length n, where $N_n=\lfloor N/n-1\rfloor$. Each segment can be written as $\varepsilon_v(i)=\varepsilon(l+i)$ for $1\le i\le n$, respectively, where $l=(v-1)n$. Then we can define the fluctuation variance as,

$$F_v^2(n)=\frac{1}{n}\sum_{i=1}^{n}\varepsilon_v(i)^2. \tag{5}$$

Step 5. Calculate the qth order fluctuation function using the equation:

$$F_q(n)=\left\{\frac{1}{n}\sum_{v=1}^{N_n}F_v^q(n)\right\}^{\frac{1}{q}}, \tag{6}$$

for $q\neq 0$. Where, $F_q(n)$ is the qth order fluctuation function and $F_v^q(n)=(F_v^2(n))^{q/2}$.
When $q=0$, according to L'Hôspital's rule, we have

$$\ln[F_0(n)]=\frac{1}{N_n}\sum_{v=1}^{N_n}\ln[F_v(n)]. \tag{7}$$

Step 6. For different values of segment length n, we have the power-law relationship,

$$F_q(n)\sim n^{h(q)}. \tag{8}$$

Here, the generalized Hurst exponent $h(q)$ can be obtained by observing the slope of log–log plot of $F_q(n)$ versus n through the method of ordinary least squares (OLS). For $q=2$, MF-DMA is just the DMA and $h(q)$ is equivalent to the seminal Hurst exponent H obtained from DMA. If $h(2)>0.5$, the auto-correlations are persistent (positive). An increase is likely to be followed by another increase. If $h(2)<0.5$, the correlations are anti-persistent (negative). An increase is likely to be followed by a decrease. If $h(2)=0.5$, the time series display a random walk behavior. The conclusion of multifractality can be obtained from the dependence of $h(q)$ on the values of fluctuation orders q. In this paper, we set the values of q to change from -10 to 10 with a fixed step of 1.

The analytical relationship between generalized Hurst exponents based on MF-DMA and Renyi exponent $\tau(q)$ is,

$$\tau(q)=qh(q)-1. \tag{9}$$

Via a Legendre transform, another important variable set $\alpha-f(\alpha)$ is defined by

$$\alpha=h(q)+qh'(q),\quad f(\alpha)=q[\alpha-h(q)]+1. \tag{10}$$

Here, α is the Holder exponent or singularity strength which characterizes the singularities in a time series. The singularity spectrum $f(\alpha)$ describes the singularity content of the time series.

According to Wang et al. (2010), an efficient market has no long-range auto-correlated behavior or multifractality. Therefore, the existence of multifractality implies the fact of market inefficiency. Moreover, the strength of multifractality can be used to quantify the degree of market inefficiency. In this sense, the investigation of multifractality and complexity has important economic implications.

4. Data

We choose daily spot price data of WTI crude oil and gasoline (New York Harbor) with the date from January 2, 1986 to July 26, 2011. The gasoline crack spreads are simply calculated using Eq. (1). Fig. 1 shows the graphical representation of the crack spread series and its first differences.

Table 1 shows the descriptive statistics of the crack spreads. The mean of crack spreads is about 6 dollars per barrel, reflecting the average marginal production cost of gasoline. The mean of the first differences is close to zero but the standard deviation is much larger. The Jarque–Bera statistics consistently show rejections of null hypothesis of Gaussian distributions at 1% significance level, indicating that crack spreads and the first differences are fat-tail distributed, also evidenced by positive skewness and excess kurtosis. The Ljung–Box statistics show that the null hypothesis of no autocorrelations up to the 10th order is rejected for both of two series, confirming the presence of serial autocorrelation.

We test for the unit root of crack spread series based on Augment Dickey and Fuller (1979) (ADF), Phillips and Perron (1988) (PP) and Kwiatkowski et al. (1992) (KPSS) methods and show the results in Table 2. To save space, we do not show the detailed description of these tests. The lag lengths of ADF test are chosen based on Schwarz information criterion (SIC) and the optimal bandwidths of PP and KPSS are determined based on Newey–West criterion. Based on each of these three methods, the unit root tests are performed by taking into account two different models (with constant and with constant and trend). The null hypothesis of ADF and PP tests is a unit root while that of KPSS test is stationarity.

The ADF and PP statistics show the rejections of null hypothesis of unit root in crack spread series at 1% significance level, indicating that they are stationary. However, KPSS statistic rejects the null hypothesis of stationarity, suggesting the opposite result from ADF and PP statistics. For the first difference series, the ADF, PP and KPSS statistics consistently support the stationarity, suggesting that crack spreads follow an I(1) process. In the following analysis, we will analyze the long-range auto-correlated behavior of the first difference series.

Brémond et al. (2011) show that after the introduction at the end of the year 2000 of the law modernizing raw materials markets-the Commodity Futures Modernization Act (CFMA), oil prices run in a new regime different from previous decade. Thus, we divide the whole time series into two sub-periods. The first period (Period I) covers the dates from January 2, 1986 to December 29, 2000 and the second period (Period II) covers the dates from January 2, 2001 to July 26, 2011. Then, we investigate the long-range auto-correlations of crack spreads in each of these two periods.

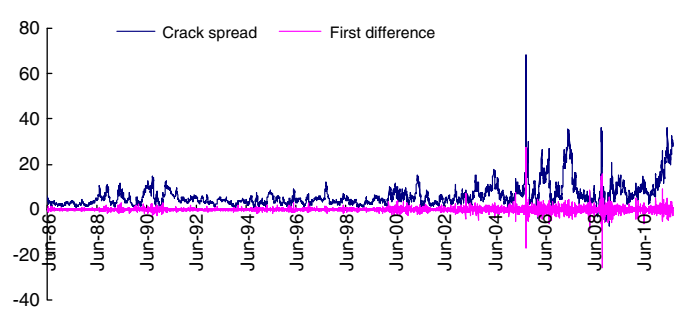

Fig. 1. Gasoline crack spreads and their first order differences.

Table 1
descriptive statistics of crack spread and the first order difference series.

	Crack spread	The first order differences
Mean	6.027	0.004
Maximum	68.500	27.530
Minimum	−8.090	−25.744
Std. Dev.	5.111	1.188
Skewness	2.674	0.559
Excess kurtosis	11.638	126.140
Jarque-Bera	43253***	4194965***
Q(10)	47959***	168.35***

Note: The Jarque-Bera statistic tests for the null hypothesis of Gaussian distribution. Q(10) is the Ljung-Box statistic of the return series for up to the 10th order serial correlation. *, ** and *** denote rejections at 10%, 5% and 1% significance level, respectively.

5. Empirical results

5.1. Long-range auto-correlations in crack spreads

In this sub-section, we will analyze the long-range auto-correlations of the crack spreads in Periods I and II. For $q=2$, MF-DMA is equivalent to DMA. Fig. 2 shows the log–log plot of detrended fluctuation function of the crack spreads in Period I, Fq(n), versus time scale, n, for $q=2$. We can find that the log–log relationships between fluctuation function and time scale cannot be characterized by one linear regime. That is, the auto-correlations of crack spreads in Period I display a multiscaling behavior, generally confirming the results on the crude oil returns in Wang and Liu (2010). The property of scaling invariance can be found in each of two intervals of time scale, $10<n<35$ and $35<n<180$. For time scales from several days to weeks ($10<n<35$), the slope of fluctuation function, just the seminal Hurst exponent is 0.4715, implying the weakly negative auto-correlated behavior in the short-term. For time scales from several months to quarters ($35<n<180$), the Hurst exponent is 0.2150 indicating that crack spreads are strongly negative auto-correlated in the long-term. In other words, both in the short-term and in the long-term, the crack spreads display mean-reverting behavior and the degree of mean-reversion is much higher in the long-term than in the short-term.

Fig. 3 shows the log–log plot of fluctuation function of crack spreads in Period II versus time scale for $q=2$. The property of scaling invariance is present for a wide range of time scales ($10<n<150$). The Hurst exponent is 0.3041, implying the strongly negative auto-correlations for the trading horizons varying from several days to quarters.

In period I, the short-term anti-persistence is very weak while the long-term anti-persistence is much stronger. In the short-term, there are multitudes of noise in petroleum markets. The short-term speculations and some external events may cause the positive auto-correlations of crude oil or gasoline prices (continuous increase or decrease) in a short period of time and lead to the persistent deviation of oil prices from long-run equilibrium level. The persistence of oil prices prevents crack spread from running toward long-run equilibrium level and results in the deviation of short-term relationships between crude oil and gasoline prices from long-term situation. Thus, the short-term auto-correlated behaviors of crack spreads may be not consistent with long-term behaviors.

Table 2
Results of unit root test.

	Crack spread		The first difference series	
	Intercept only	Intercept and trend	Intercept only	Intercept and trend
ADF	−6.204***	−7.209***	−37.749***	−37.751***
PP	−8.018***	−9.319***	−91.830***	−91.959***
KPSS	2.812***	0.326***	0.017	0.023

Note: This table reports the results of testing for unit root in crack spreads and their first order differences. The null hypothesis of ADF and PP test is unit root and that of KPSS test is stationarity. The optimal lag lengths of ADF test are chosen based on Schwarz information criterion (SIC) and the optimal bandwidths of PP and KPSS are determined based on Newey–West criterion. *, ** and *** denote rejections of null hypothesis at 10%, 5% and 1% significance level, respectively.

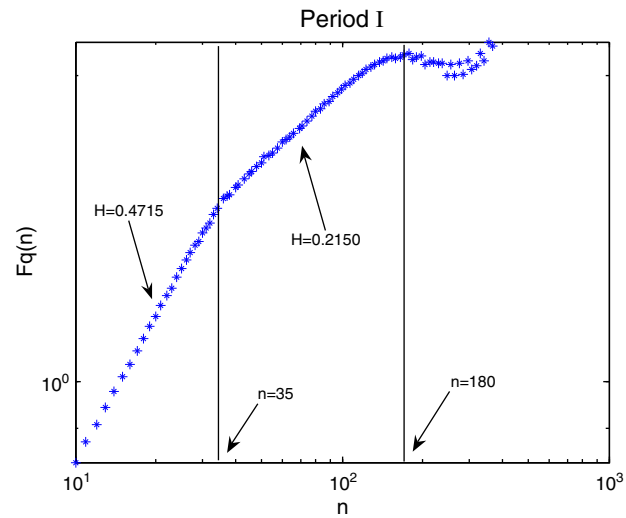

Fig. 2. Log–log plot of fluctuation function Fq(n) of crack spreads in Period I versus time scale n for q = 2.

The long-term mean reversion is due to the fact that crack spread is a representation of a profit margin (the margin for refining crude oil into gasoline) (Dunis et al., 2006b). The crack spread measure used in this paper implies that the refining ratio for crude oil is 1:1:0.[2] That is, 1 part crude can be refined into 1 part gasoline. In fact, besides gasoline, there are a large range of crude oil products (heating oil, jet fuel, fuel oil, propane, etc.), which take a considerable ratio of oil products. The volatile prices of these products play significant roles in the relationships between crude oil and gasoline prices. The fluctuating demand of these products results in the fact that the gasoline crack spread may move away from the long-run equilibrium level. The crack spread can reflect major information about the refining cost. With the development of technology, refining technology has been reached an advanced level. The long-term mean-reversion of crack spreads also attributes to the almost stagnant refining technology.

Fig. 4 shows the multifractal spectra of crack spreads via Legendre transform. The multifractal spectra display parabola shapes, suggesting the multifractality of long-range auto-correlations. The left part of multifractal spectrum related to positive values of q describes the scaling behaviors of large fluctuations. The right part of spectrum related to negative values of q describes the scaling behaviors of small fluctuations. Thus, the multifractality implies that small and large fluctuations display different scaling behaviors. The width of spectrum of crack spreads in Period II is larger than that in Period I, indicating that the degree of multifractality in Period II is higher than that in Period I. As shown in Alvarez-Ramirez et al. (2008) and Wang and Liu (2010), the degree multifractality can be used to quantify the market complexity. Thus, we can say that the crack spreads in Period II have a higher degree of complexity. According to Wang et al. (2010), multifractality also implies market inefficiency. In this sense, crude oil market is more inefficient in Period II than in Period I. Additionally, we find an abnormal phenomenon that negative fractal dimensions ($f(\alpha)<0$) emerge in the multifractal spectra,

[2] For convenience, we investigate the crack spreads with the ratio 1:1:0. We also investigate the crack spreads with another more common ratio 3:2:1 and obtain very similar results to the crack spreads with the ratio 1:1:0.

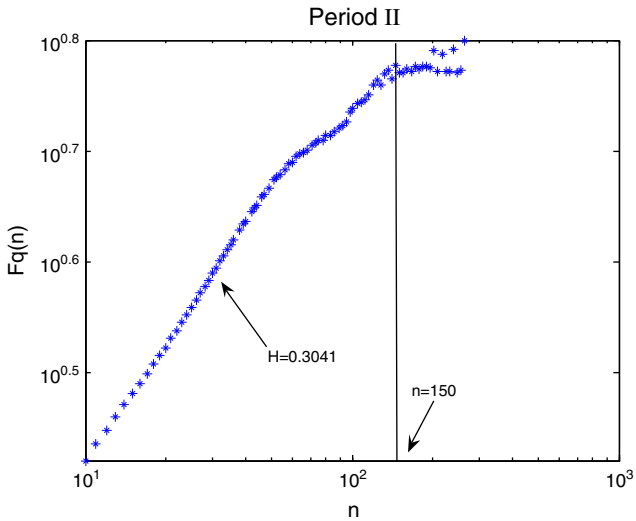

Fig. 3. Log–log plot of fluctuation function Fq(n) of crack spreads in Period II versus time scale n for q = 2.

especially for the crack spreads in Period II. Most of the negative fractal dimensions appear in the left part of multifractal spectra, indicating that the major source of negative fractal dimensions is some large market fluctuations. This evidence confirms that some extreme events such as recent financial crisis and the second Gulf War have great effects on energy market dynamics, such as long-range auto-correlations of crack spreads. More negative fractal dimensions in Period II also indicate that the higher degree of market inefficiency in this period is partly due to the more frequently large fluctuations.

To study the evolution of long-range auto-correlations over time, following Tabak and Cajueiro (2007) and Wang and Liu (2010), we use the technique of "rolling windows" (rolling sample test). The length of each window is fixed to be 1000 business days (about 4 years). We first examine the long-range auto-correlations of crack spreads in the first window, covering the period from January 2, 1986 to May 21, 1990. Then, the estimation period rolls forward by adding a new business day and dropping the most distant business day. With the window rolls forward, the evolution of long-range auto-correlated behavior is obtained.

Fig. 5 shows the graphical representation of the dynamics of Hurst exponents over time. The date in *x*-axis stands for the mid-point of

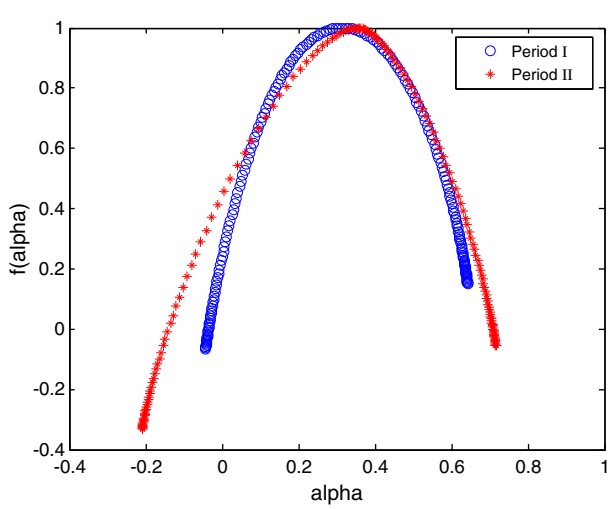

Fig. 4. Multifractal spectra of crack spreads in two periods.

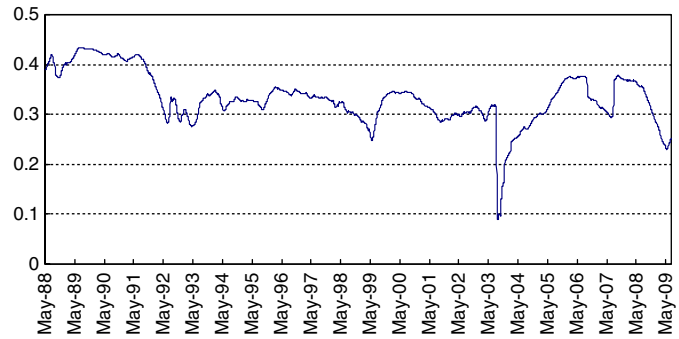

Fig. 5. Time-varying Hurst exponents of crack spreads.

the sample in each window. Overall, with window rolls forward, the Hurst exponents keep varying under the line of 0.5, which further confirms the anti-persistent pattern of long-range auto-correlations obtained from former analysis. That is also to say, the long-range auto-correlations in crack spreads are time-varying, consistent with the findings on crude oil returns in existing literature (Alvarez-Ramirez et al., 2008; Alvarez-Ramirez et al., 2010; Tabak and Cajueiro, 2007; Wang and Liu, 2010).

In Fig. 5, we can find the effects of several extreme events on long-range auto-correlated behavior of crack spreads. From May 1988 to the end of 1990, the Hurst exponent varies around 0.4. Then, a crash of Hurst exponent is found in the period 1991–1992, related to the effects of first Gulf War. We can also find several crashes in the period mid-1997–1999, at the end of 2003 and in the period 2008–2009, which can be related to the Asian financial crisis, the second Gulf War and the recent financial crisis, respectively. Overall, these major events may cause crashes of the Hurst exponent, strengthening the degree of anti-persistence. However, we should note that the effects of these events on long-range auto-correlations are different, further confirming the results in Kilian (2010) that not all oil shocks are alike. For example, after the crash which occurred in the period of 1991–1992, the Hurst exponents varies around 0.3 for several years, lower than those before the crash. In contrast to the first Gulf War, after the crash which occurred in 2003, the Hurst exponents quickly go back to the pre-crash level. That is, the effect of the first Gulf War on long-range auto-correlations is permanent but the effect of the second Gulf War is temporary. In Fig. 5, we can also find that the effect of the Asian financial crisis is temporary. For the lack of enough data period, whether the effect of the recent financial crisis is permanent or temporary is ambiguous.

We quantify the degree of multifractality using the width of multifractal spectra, which is the range of singularity exponent α, $\Delta\alpha = \alpha_{max} - \alpha_{min}$. Fig. 6 shows the evolution of multifractality degree with window rolling forward. Overall, the multifractality

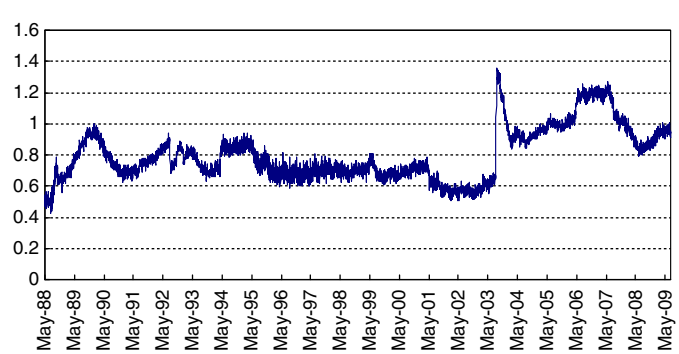

Fig. 6. Time-varying degrees of multifractality of crack spreads.

degrees are time-varying, and keep relatively high values after 2003. We can clearly find that some major events such as two Gulf Wars and recent financial crisis can drive the degrees of multifractality to rise up in a short period of time, especially the second Gulf War which occurred in 2003. From the dynamics of multifractality degrees, we can say that some extreme events can cause the high degree of market complexity. In other words, all of these events can lead to more inefficient market operations.

Another important issue which should be addressed is whether the volatility of crack spreads are long-range auto-correlated. We use absolute difference as the proxy of volatility following Tabak and Cajueiro (2007) and Wang et al. (2011a, 2011b). We present a graph of Hurst exponents for volatility with window rolling forward. The Hurst exponents change round 1 and are less volatile than those of crack spreads (see Fig. 7). That is, the volatility is strongly long-range auto-correlated, almost permanent persistence. Although there is a large crash in 2003, the Hurst exponents come back to 1 after a very short period of time. Thus, we can say that the effect of the second Gulf War on long-range auto-correlations of crack spread volatility is temporary.

Fig. 8 shows the graphical representation of the evolution of multifractality degrees. The degrees of multifractality change over time, implying the time-varying complexity of crack spread volatility. After 2003, the degrees of multifractality jump to relatively high levels and then gradually decrease over time. Recently, the degree of multifractality seems to maintain the level of 0.8.

5.2. Modeling and forecasting implications

Our results have indicated that the long-range auto-correlations of crack spreads display multiscaling behavior, mean-reversion, and a high degree of multifractality (complexity). Modeling and forecasting the crack spread are of great importance for market participants. Thus, whether some conventional models can capture these dynamics of crack spread deserves further investigate.

We employ an autoregressive fractionally integrated moving average (ARFIMA) to model the mean of crack spread. The ARFIMA (1,d,1) model with the mean μ for crack spreads can be given by

$$(1-\varphi)(1-L)^d(\Delta CS-\mu) = (1+\theta L)\varepsilon_t \tag{11}$$

where, ΔCS_t is the first order difference of crack spread at day t, which has been proved to be stationary. L is the lag operator coefficients d, φ and θ are fixed and unknown parameters and ε_t is Gaussian white noise with mean zero and variance σ_ε^2. Since different ARFIMA model specifications produce rather similar forecasting results (Koopman et al., 2005), we only consider ARFIMA(1,d,1) model in the following empirical analysis.

As the auto-correlations of crack spread volatility are strongly persistent, we use generalized autoregressive conditional heteroskedasticity

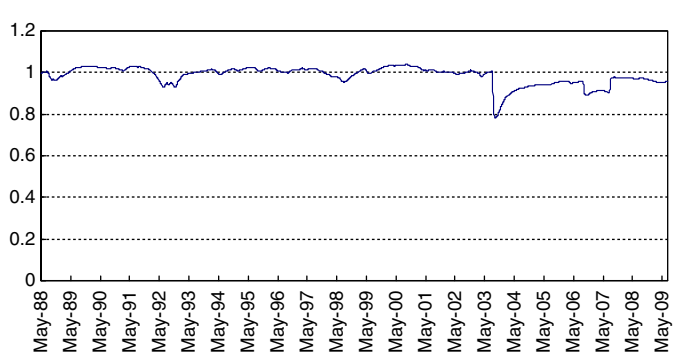

Fig. 7. Time-varying Hurst exponents of crack spread volatilities.

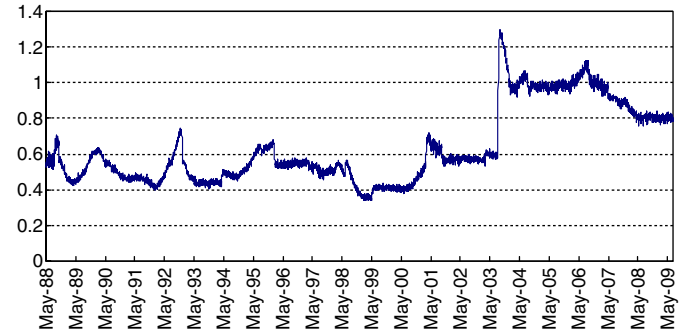

Fig. 8. Time-varying degrees of multifractality of crack spread volatilities.

(GARCH) (Bollerslev, 1986) models to capture the dynamics of crack spread volatility. We consider the four GARCH-class models, GARCH(1,1), GJR, FIGARCH and IGARCH models, which can be described as follows.

5.2.1. GARCH (1,1) model

An important extension work of autoregressive conditional heteroskedasticity (ARCH) in Engle (1982) is generalized ARCH (GARCH) introduced by Bollerslev (1986). Bollerslev et al. (1994) also showed that GARCH (1,1) model performed well in most situations. Evidence from Sadorsky (2006) also showed that GARCH (1,1) model fit very well for volatility of crude oil prices. The GARCH (1,1) model can be described as follows:

$$\begin{aligned} &\Delta CS_t = \mu_t + \varepsilon_t = \mu_t + \sigma_t z_t, \quad z_t \sim NID(0,1), \\ &\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \end{aligned} \tag{12}$$

where μ_t denotes the conditional mean and σ_t^2 the conditional variance with parameter restrictions $\omega>0$, $\alpha\geq0$, $\beta\geq0$ and $\alpha+\beta<1$.

Another linear GARCH-type model is IGARCH model of Engle and Bollerslev (1986), which can captures infinite persistence in the conditional variance. The model setting of IGARCH(1,1) model is similar to GARCH(1,1) but with parameter restrictions $\alpha+\beta=1$.

5.2.2. GJR-ARCH model

GJR-ARCH model was proposed by Glosten et al. (1993) to capture the asymmetric volatility effect by adding a binary variable to the variance equation. The variance equation can be described as follows:

$$\sigma_t^2 = \omega + [\alpha + \gamma I(\varepsilon_{t-1}<0)]\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{13}$$

where $I(.)$ is an indicator function, i.e., when condition in ( ) is met, $I(.)=1$ and 0 otherwise. γ is the asymmetric coefficient to describe volatility asymmetric effect.

5.2.3. FIGARCH model

Some scholars claimed that long memory properties in volatility of financial markets could be better captured by FIGARCH model proposed by Baillie et al. (1996). The FIGARCH model implied that the autocorrelations in volatility decay at a slow hyperbolic rate, not at an exponential rate as described in GARCH and EGARCH models. The specific form of FIGARCH (1,d,1) model which is the most prevalent model is shown as follows:

$$\sigma_t^2 = \omega + \beta\sigma_{t-1}^2 + \left[1-(1-\beta L)^{-1}(1-\varphi L)(1-L)^d\right]\varepsilon_t^2 \tag{14}$$

where $0\leq d\leq1$, $\omega>0$, φ, $\beta<1$. d is the fractional integration parameter and L the lag operator. The parameter d characterizes the long memory property of hyperbolic decay in volatility. The appreciation of the FIGARCH process is that for $0<d<1$, it is sufficiently

flexible to allow for intermediate ranges of persistence, between complete integrated persistence of volatility shocks associated with $d=1$ (IGARCH,) and the geometric decay associated with $d=0$ (GARCH(1,1)).

To evaluate the model performances, we compare the scaling behavior of crack spreads and their GARCH standardized residuals which can be described using the equation as follows:

$$\xi_t = \varepsilon_t/\sigma_t, \tag{15}$$

where ξ_t is GARCH standardized crack spread at day t, ε_t is residual in Eq. (11) and σ_t is the crack spread volatility captured by GARCH-class models.

If ARFIMA-GARCH models can well capture the dynamics of crack spreads, the standardized residuals in Eq. (15) should be a white noise, implying the monoscaling behavior, no long-range auto-correlations ($H=0.5$) and zero multifractality degree. Figs. 9 and 10 provide the log–log plots of fluctuation function (Fq(n)) of standardized residuals in Period I and Period II versus time scale n for q = 2, respectively. The corresponding Hurst exponents are shown in Tables 3 and 4. In Fig. 9, four GARCH standardized residuals display similar scaling behaviors. A crossover of log–log relationship between fluctuation function and time scale at about n = 35 is present, implying the similar multiscaling behavior to the long-range auto-correlations of original crack spreads. The Hurst exponents are closer to 0.5 than those of original series but not equal to 0.5, indicating that ARFIMA-GARCH models can partly capture the long-range auto-correlations of crack spreads, also confirmed by the Hurst exponents of standardized residuals in Period II (see Table 4 and Fig. 10).

Figs. 11 and 12 provide the multifractal spectra of standardized residuals. The widths of spectra of standardized residuals are much narrower than those of original series, indicating that ARFIMA-GARCH models can partly capture the multifractality (complexity) of crack spreads, also confirmed by the multifractality degrees of shown in Tables 3 and 4. Moreover, we find that after the ARFIMA-GARCH filtering, the left parts of multifractal spectra shrink greatly while the right parts almost have no change and the negative fractal dimensions in the left parts almost disappear. That is, ARFIMA-GARCH models can capture the complexity of crack spreads caused by large fluctuations. For the dynamics of small fluctuations, ARFIMA-GARCH models are misspecified.

Our results show that the crack spreads are long-range auto-correlated and the specific form of the long-range auto-correlations is anti-persistence. It implies that ARFIMA model which takes the long-range auto-correlations into account models the crack spreads better than random walk model (RWM). However, whether this implication still holds for out-of-sample situation is not clear. For this consideration, we compare the forecasting ability of ARFIMA (1,d,1) and RWM.

The forecasting procedure is based on the technique of "rolling windows" described in previous analysis. The whole sample with 6327 data points is classified into two subgroups: 1) in-sample data for modeling, covering the first 1000 trading days, and 2) out-of-sample data for model evaluation, covering the last 5327 trading days of the total data sample. The estimation period is then rolled forward by adding one new day and dropping the most distant day in each window. In this way, the length of each rolling window is fixed and the forecasts do not overlap, thereby allowing daily (one-day-ahead) out-of-sample forecasts to be obtained.

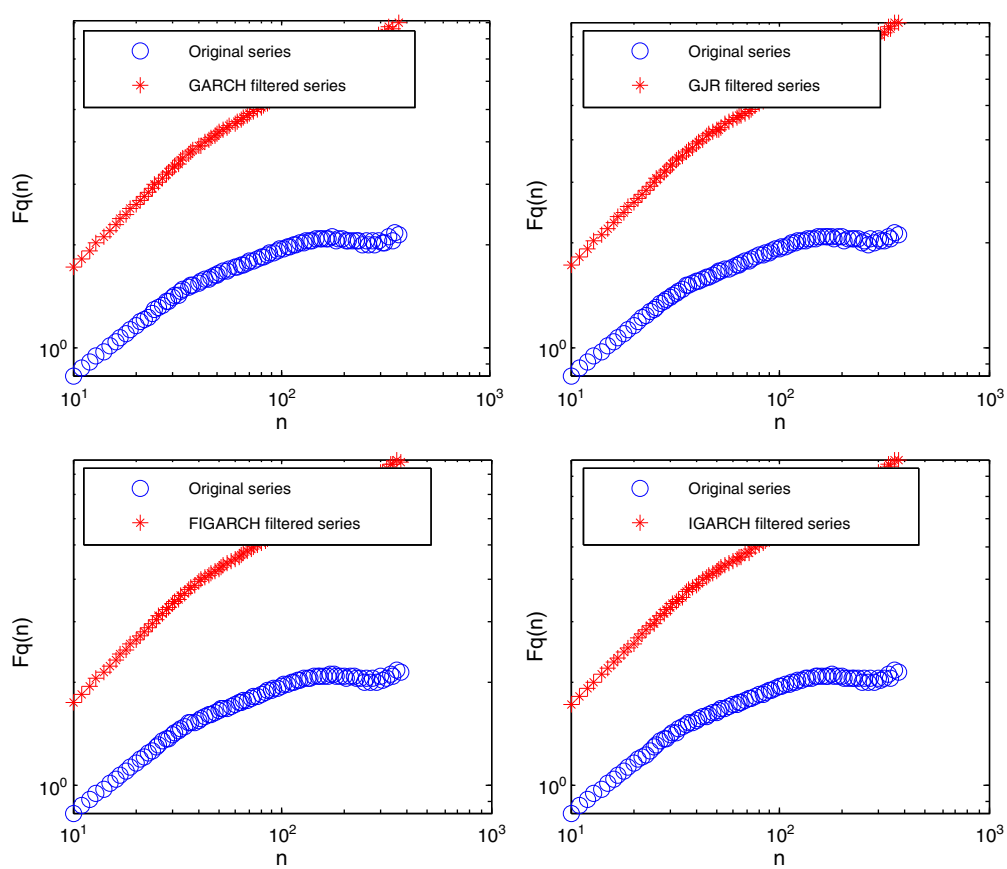

Fig. 9. Log–log plots of fluctuation function Fq(n) versus time scale n for crack spreads and model filtered residuals in Period I.

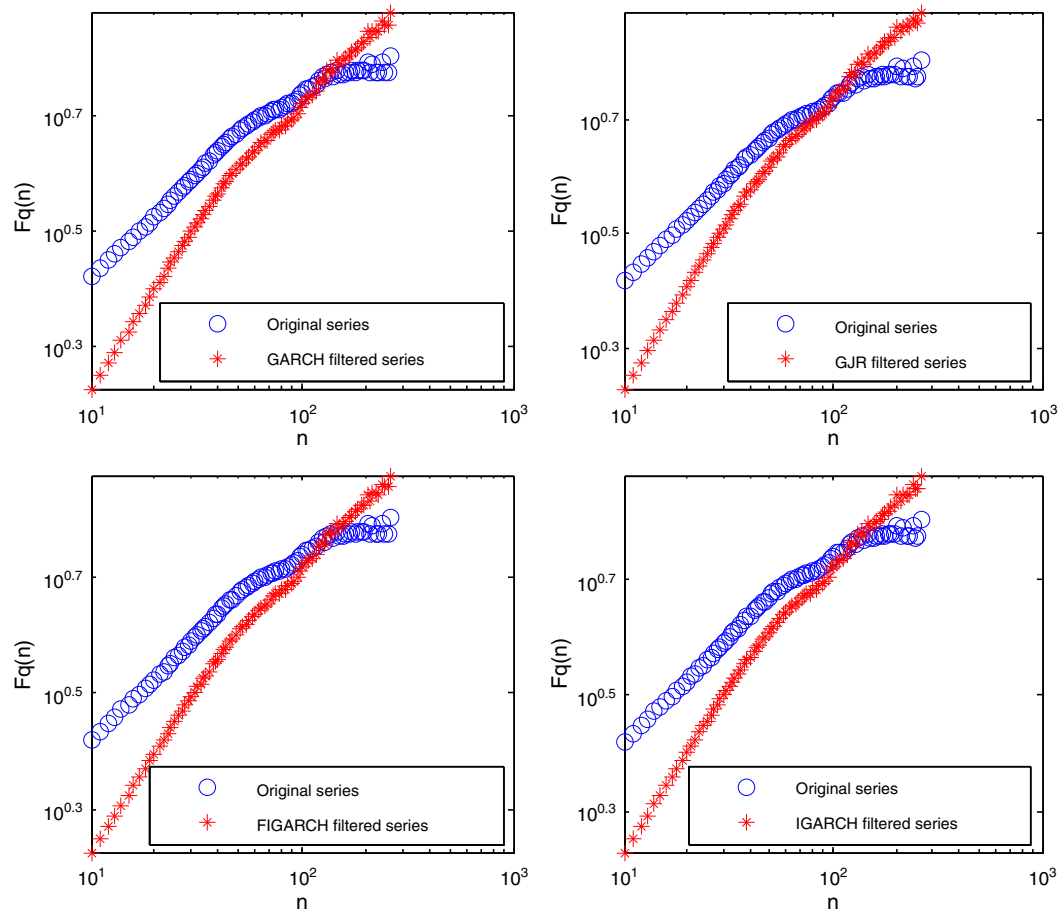

Fig. 10. Log–log plots of fluctuation function Fq(n) versus time scale n for crack spreads and model filtered residuals in Period II.

To measure the forecast accuracy, we take the mean of squared errors (MSE) and the mean of absolute errors (MAE) of the forecasts as the criterions,

$$\mathrm{MSE} = n^{-1}\sum_{t=1}^{n}\left(CS_t - \hat{CS}_t\right)^2, \tag{16}$$

$$\mathrm{MAE} = n^{-1}\sum_{t=1}^{n}\left|CS_t - \hat{CS}_t\right|, \tag{17}$$

where, n is the number of forecasting data points, $\hat{CS}_t$ is the forecasting crack spread implied by the selected models.

Although the above statistics of forecast errors, just the loss functions, can be used to well evaluate the performance of models, they do not provide the statistical tests of the difference between two models. For this purpose, we take statistics proposed by Diebold and Mariano (1995). The DM test can be described as follows:

For n daily forecasts from two different models, the forecast errors are denoted as $e_{1,t}$ and $e_{2,t}$, where t = 1,2,…,n. the hypothesis of equal forecast accuracy can be represented as $E(d_t) = 0$, where $d_t = g(e_{1,t}) - g(e_{2,t})$ and E is the expectation operator, and $g(e_{1,t})$ and $g(e_{1,t})$ denote a kind of loss function of forecast for two selected models. The mean of the difference of the forecast errors between two models $\bar{d} = n^{-1}\sum_{t=1}^{n} d_t$ has an approximate asymptotic variance of,

$$V(\bar{d}) \approx n^{-1}\left[\lambda_0 + 2\sum_{k=1}^{h-1}\lambda_k\right], \tag{18}$$

where λ_k is k-th auto-covariance of d_t, which can be estimated as,

$$\hat{\lambda} = n^{-1}\sum_{t=k+1}^{n}(d_t - \bar{d})(d_{t-k} - \bar{d}). \tag{19}$$

Table 3
The statistics of the long-range auto-correlations of crack spreads in Period I.

	Hurst exponent (10<n<35)	Hurst exponent (35<n<150)	Multifractality degree
Original series	0.4715	0.2150	0.6853
GARCH(1,1) standardized residuals	0.5276	0.3893	0.2429
GJR standardized residuals	0.5289	0.3844	0.2508
FIGARCH standardized residuals	0.5268	0.3850	0.2538
IGARCH standardized residuals	0.5272	0.3877	0.2397

Table 4
The statistics of the long-range auto-correlations of crack spreads in Period II.

	Hurst exponent (10<n<180)	Multifractality degree
Original series	0.3041	0.9257
GARCH(1,1) standardized residuals	0.4636	0.4472
GJR standardized residuals	0.4774	0.4978
FIGARCH standardized residuals	0.4636	0.4925
IGARCH standardized residuals	0.4631	0.4496

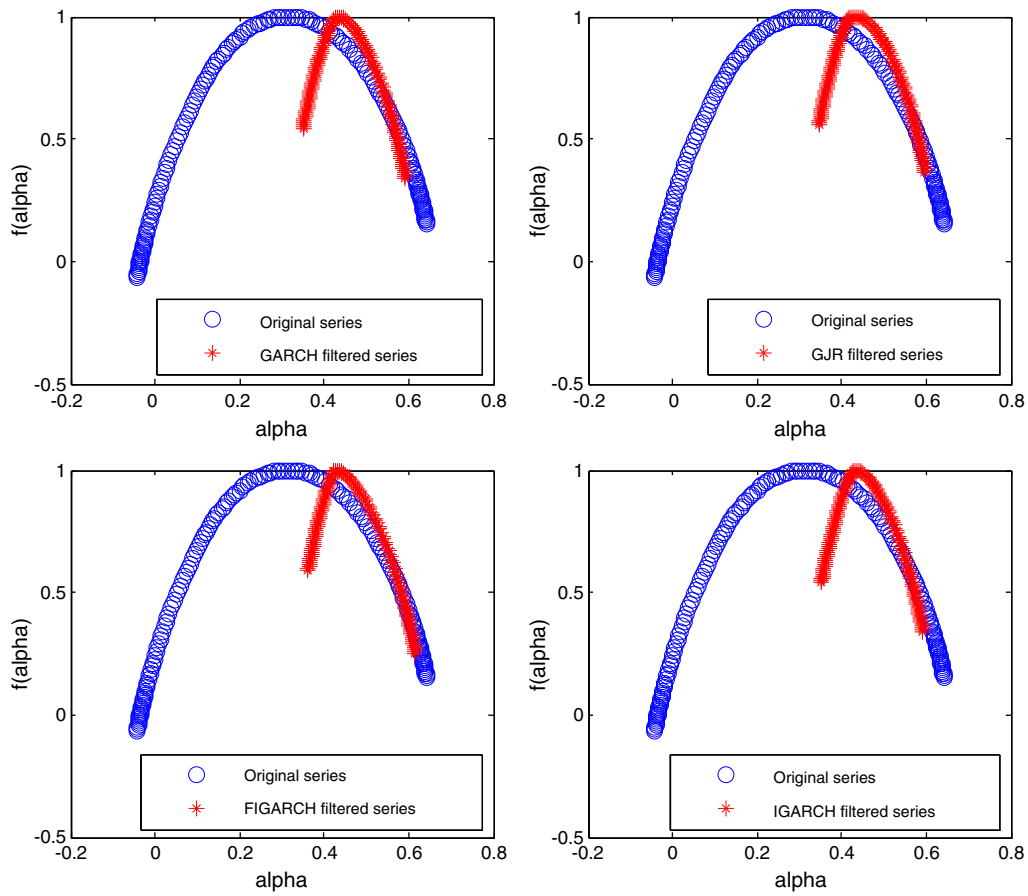

Fig. 11. Multifractal spectra of crack spreads and model filtered residuals in Period I.

The DM test for the null hypothesis of equal forecast accuracy is,

$$DM = \left[V(\hat{\bar{d}})\right]^{-1/2} \bar{d}, \tag{20}$$

where DM has an asymptotic standard normal distribution under the null hypothesis. In our case, the DM test is calculated from the loss difference based on the MSE and MAE criteria of different forecasting models.

Table 5 provides the results of daily forecasting accuracy of RWM and ARFIMA (1,d,1) model measured by two loss functions, MAE and MSE. We also examine the null hypothesis of no difference in accuracy between ARFIMA and RWM. The RWM is chosen as the base model. Under the criteria of MSE and MAE, the loss functions of ARFIMA are larger than those of RWM, at least at 10% significance level, indicating that RWM has a higher degree of forecasting accuracy. That is, the strong long-range auto-correlations imply that ARFIMA(1,d,1) models the dynamics of crack spread better than RWM, but it does not imply that ARFIMA outperforms RWM in the sense of out-of-sample prediction.

6. Conclusion

Crack spread, the price differential between crude oil and its refined products, provides important information on the marginal production costs. Oil refiners and consumers are frequently exposed to the crack spreads. Speculators and hedge funds are heavily using crack spread derivatives to speculate in petroleum markets. Thus, the dynamics of crack spreads are of great importance for market participants.

In this paper, we investigate the long-range auto-correlations of gasoline crack spreads. We divide the whole sample period into two parts, Period I with relatively low variability of crack spreads (January 2, 1986–December 29, 2000) and Period II with relatively high variability of crack spreads (January 2, 2001–July 26, 2011). Using a multifractal detrended moving average (MF-DMA) (Gu and Zhou, 2010), which is superior to multifractal detrended fluctuation analysis (MF-DFA) employed in Alvarez-Ramirez et al. (2008) and Wang and Liu (2010), we find the long-range auto-correlations of crack spreads in Period I display a multiscaling behavior, generally consistent with Wang and Liu (2010). Specifically, for the time scales from days to weeks, the crack spreads display weakly anti-persistent behaviors while for the time scales from weeks to months, the anti-persistence is much stronger. For the crack spreads in Period II, the long-range auto-correlations are also anti-persistence with time scales varying from days to months, implying the mean-reverting behaviors. Our results are consistent with the results on energy price returns in Serletis and Rosenberg (2007) and Elder and Serletis (2008). Moreover, based on multifractal analysis, we find that the long-range auto-correlations of crack spreads are multifractal in both of two periods. That is, various small and large fluctuations of crack spreads have different scaling behaviors. The existence of multifractality also implies the inefficiency of crude oil market in Fama (1970)'s sense.

For the purpose of investigating the dynamics of long-range auto-correlations over time, we employ a technique of rolling windows (Alvarez-Ramirez et al., 2008; Tabak and Cajueiro, 2007; Wang and Liu, 2010). Our results show that although the long-range auto-correlations are time-varying, the specific form of long-range auto-correlations is anti-persistence and has no change. Some supply and demand shocks can cause the crashes of Hurst exponents and the

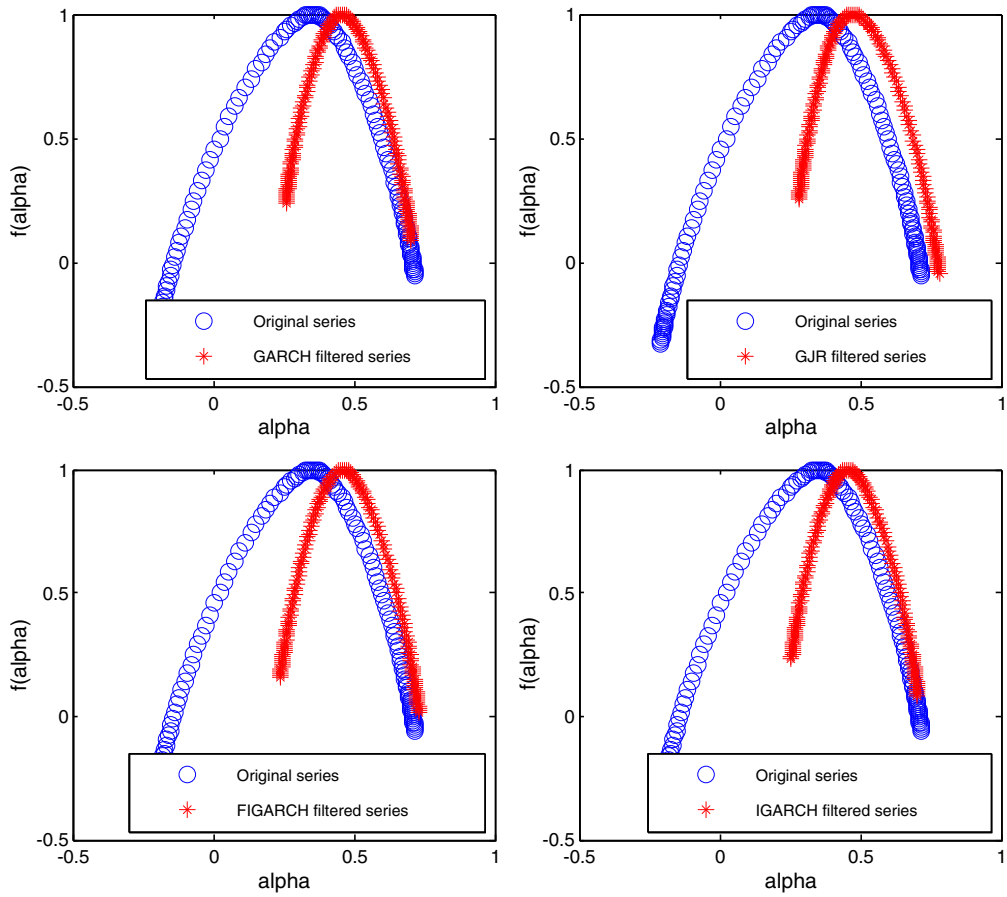

Fig. 12. Multifractal spectra of crack spreads and model filtered residuals in Period II.

increase of the degree of anti-persistence. However, not the shocks of all the events on long-range auto-correlations are alike. The effects of some events on crack spreads are temporary such as the Asia financial crisis in 1997–1998 and the second Gulf War in 2003 and the effects of some other events are permanent, such as the first Gulf War in 1991–1992. This result is consistent with Kilian (2010) who claims that not all of the crude oil shocks are alike. Additionally, the evidence based on rolling windows indicates that the degrees of multifractality which can be used to quantify the complexity are time-varying. Some extreme events can drive the degrees of complexity to rise up and thereby have negative effects on oil market efficiency.

We also investigate the long-range auto-correlations of crack spread volatilities. We find that the long-range auto-correlations of volatility are strongly persistent and less volatile than those of crack spreads. The degrees of complexity are time-varying and seem to become higher recently.

We discuss the model implications of our findings. The results indicate that ARFIMA-GARCH models can only partly capture the long-range auto-correlations and multifractality (complexity) of crack spreads. Moreover, ARFIMA-GARCH models can well capture the dynamics of large fluctuations. However, for small fluctuations, the ARFIMA-GARCH models are misspecified. Interestingly, we find that although crack spreads are strongly long-range auto-correlated, ARFIMA cannot outperform random walk model (RWM) in the sense of out-of-sample prediction. The major reason may be that the high degrees of complexity result in the larger forecasting loss of ARFIMA model than RWM.

We have investigated the long-range auto-correlations of crack spreads. Our findings also have some other implications. For example, the mean-reverting behavior of crack spreads also implies that the long-term stable equilibrium relationships between crude oil and gasoline prices, supporting the energy market integration. The time-varying anti-persistence of long-range auto-correlations implies that the degree of market integration is time-varying.

We investigate the crack spreads with the ratio 1:1:0. For robustness, we also investigate the crack spreads with another more common ratio 3:2:1 and obtain very similar results. We hope that our results can provide more useful information to oil market participants and researchers.

Table 5
The comparison of forecasting accuracy.

	MAE	MSE
RWM	0.915	3.177
ARFIMA(1,d,1)	0.946***(6.866)	3.781*(1.864)

Note: This table provides the loss functions of forecasting models. The numbers in parentheses are DM statistics. *, ** and *** denote rejections of null hypothesis at 10%, 5% and 1% significance level, respectively.

Acknowledgment

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